MID-STATE TECHNICAL COLLEGE · IT PROGRAMS

Configuring Initial Firewall Settings

Unit 2 · Information Security 2 (10-151-111)

MID-STATE TECHNICAL COLLEGE

Learning Outcomes

By the end of this unit, you will be able to:

Identify management interfaces

Name the paths into a firewall — MGT, console, web, CLI, API, Panorama — and when each applies.

Configure management network settings

Set the MGT address, gateway, permitted IP addresses, and enabled services, then commit.

Enable platform services

Configure DNS and NTP so the firewall can resolve, timestamp, and reach the update servers.

Manage licenses and software

Explain where licenses live and how PAN-OS and dynamic content updates are applied.

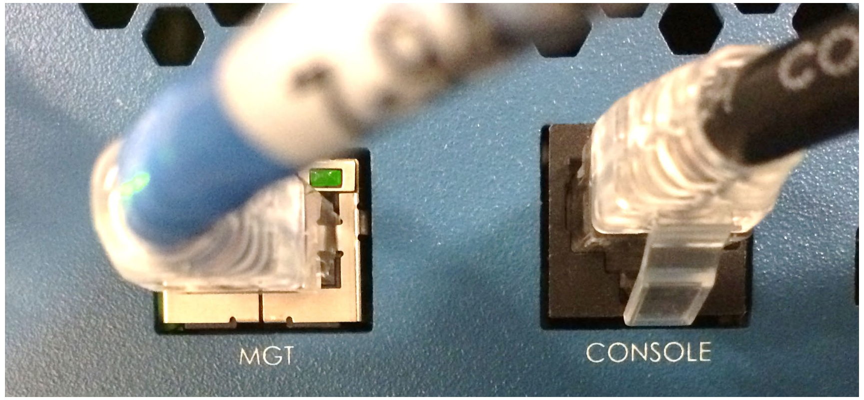
LESSON 1

Initial System Access

Reaching a New Firewall

Initial configuration happens over a path that does not depend on the firewall working yet.

- **MGT interface** — a dedicated out-of-band management Ethernet port, separate from every data plane interface.
- **Serial console** — the fallback when the network is unavailable or the MGT address is unknown.
- **Default addressing** — most hardware models come up on `192.168.1.1/24`; VM-Series comes up as a DHCP client.
- **Predefined administrator** — `admin` / `admin`, which is exactly why changing it is step one.



Two ports, two paths in. Learn where both are before you need the second one.

Administrative Access Tools

Once you can reach the box, four interfaces are available — each with a different job.

Interface	What it is	Use it for
Web interface	Graphical administration over HTTPS	Configuration and monitoring — our default
SSH / console CLI	Operational and configure command modes	Repetitive work, troubleshooting, recovery
REST / XML API	Programmatic access	Automation and integration
Panorama	Centralized management across firewalls	Device groups and templates at scale

A few recovery tasks are console-only. That is not a preference — it is the only path that works.

Reset to Factory Configuration

Password known

- Run `request system private-data-reset` from the CLI
- Erases all logs and resets every setting
- IP addressing resets too — you will lose connectivity
- Saves a default configuration once MGT changes

Password unknown

- Connect to the console port
- Type `maint` during bootup
- Choose Reset to Factory Default
- No network path exists — you must be at the device

A middle option: If the admin password changed but the box is still reachable, loading a previous configuration file is often the faster recovery.

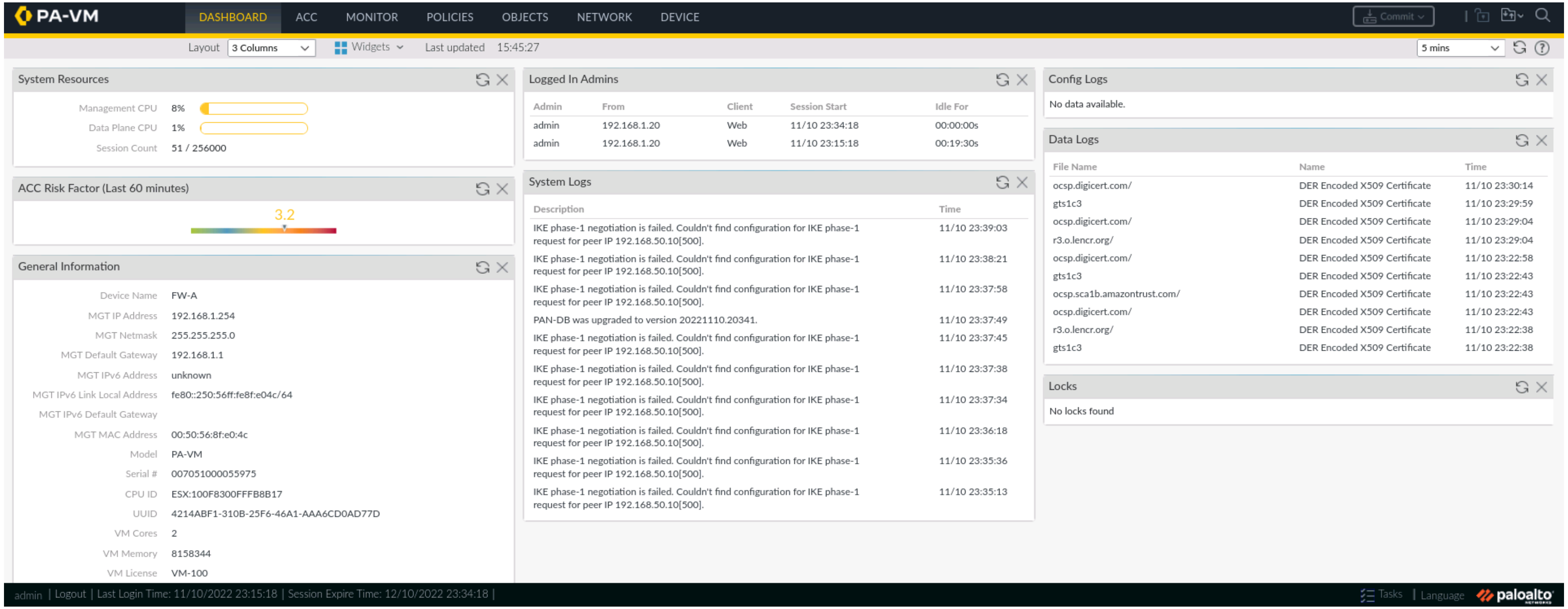
Physical access is the last resort and the last defense. Both facts matter.

LESSON 2

The Web Interface and Commit

Touring the Web Interface

Seven functional tabs across the top; almost everything in this unit lives under Device.



Commit sits at the top right. Tasks sits at the bottom right — that is how you confirm a job actually finished.

Candidate vs Running Configuration

Candidate configuration

- Your in-progress edits
- Visible in the interface, saved across the session
- **Not enforced on any traffic**
- Discardable — revert to running or last saved

Running configuration

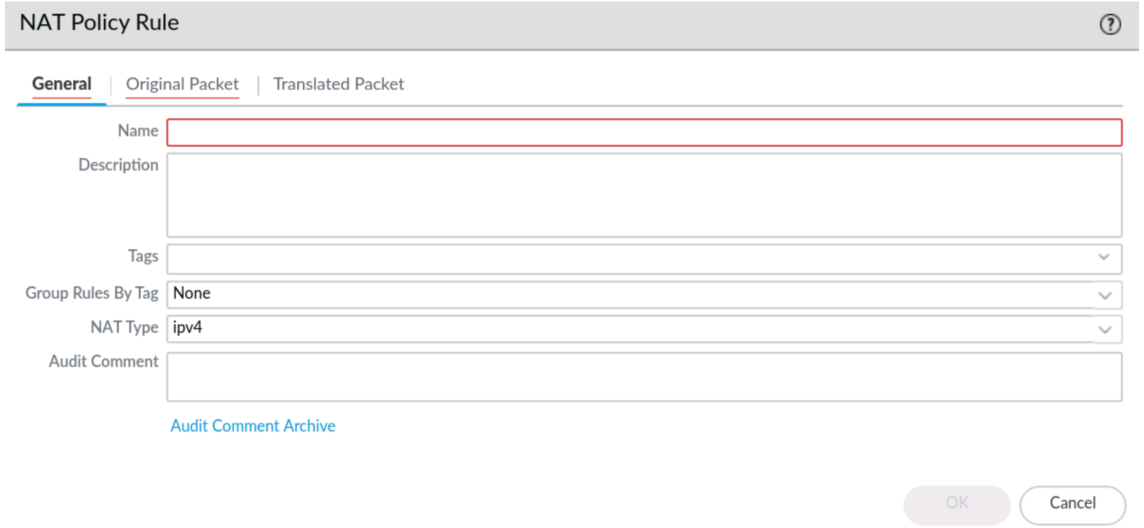
- What the firewall is actually doing right now
- Replaced only by a commit
- Every change since the last commit activates together
- Backed up as a named configuration snapshot

Before you commit: The Commit window offers Preview Changes, Change Summary, and Validate Commit. On a production firewall, use all three.

LIVE DEMO *Change a field, show it unenforced, then commit and show the job in Task Manager.*

Reading the Interface

- A **red-outlined field** is required. `OK` stays disabled until every one is filled.
- Tabs inside a dialog hold more required fields — check all of them before hunting for the problem.
- **Closing a dialog** saves to the candidate configuration only. It commits nothing.
- Device > Setup > Operations holds save, load, revert, and named snapshots.



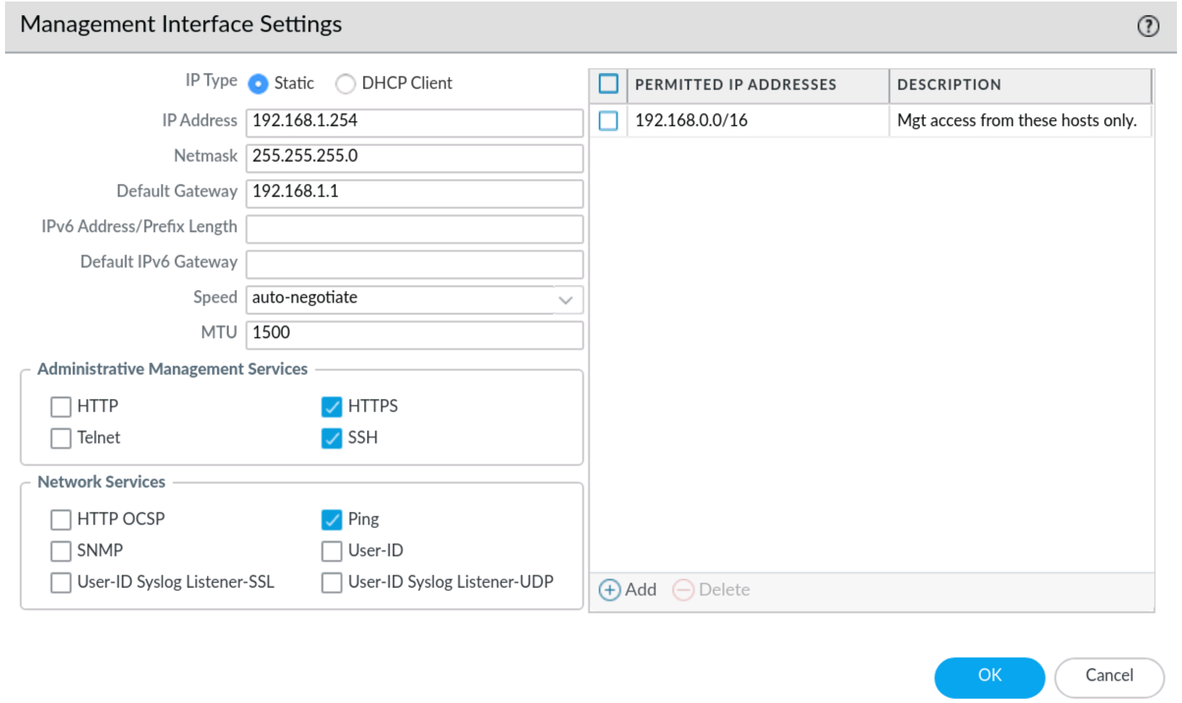
LESSON 3

Management Network Settings

MGT Interface Configuration

Device > Setup > Interfaces > Management

- **Static or DHCP** — static in practice, because an admin address that moves is an admin address you cannot reach.
- **Administrative services** — enable `HTTPS` and `SSH`. Never `HTTP` or `Telnet`.
- **Network services** — Ping is useful; SNMP and the User-ID listeners only if you use them.
- **Permitted IP addresses** — an empty list means anywhere.



Permitted IP Addresses

- An **empty list** means the management interface accepts connections from any reachable address.
- Populate it with the specific hosts or subnets that administer this firewall.
- A `/32` entry permits one host. A `/24` permits 254 of them — read the mask, not the description.
- Ideally the only entries are your **bastion or jump hosts**.

SECURITY FOCUS

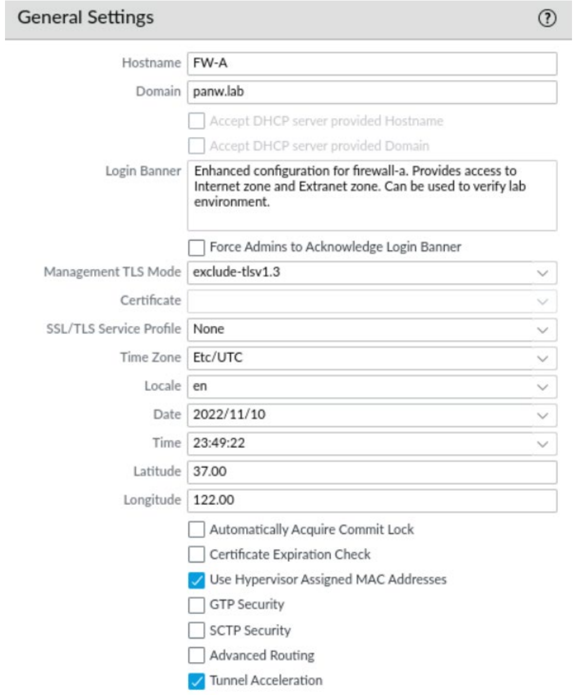
This one list is what limits the blast radius when administrator credentials are stolen or an authentication flaw is found.

Why it matters: Stolen credentials are worth nothing to an attacker who cannot reach the login page.

General Settings

Device > Setup > Management — the settings that shape how the firewall reports and how admins are greeted.

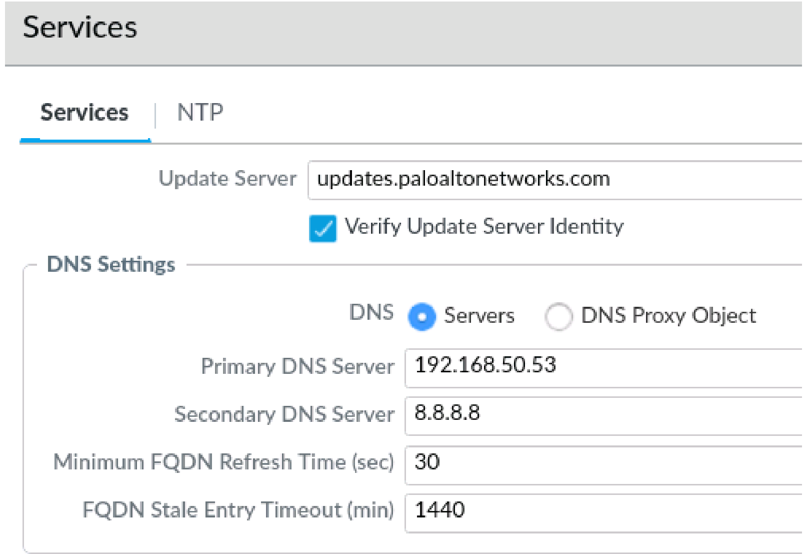
- **Hostname and domain** — both default to the model name. Set them before logs from several firewalls share a destination.
- **Login banner** — optional, but standard practice, and it can require acknowledgment before login proceeds.
- **Time zone** — every log entry, scheduled update, and cross-system correlation depends on this.
- **Latitude and longitude** — places the firewall on the ACC and Monitor maps.

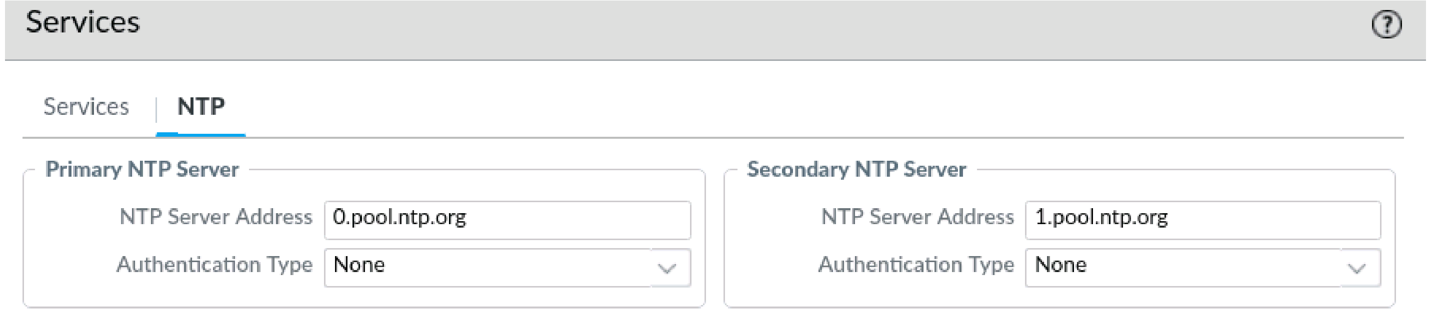


DNS and NTP

Device > Setup > Services — the platform services everything else depends on.

- **DNS servers** — required. Used for FQDN address objects, logging, and reaching the update server.
- **Update server** — defaults to `updates.paloaltonetworks.com` with identity verification on. Leave both alone.
- **NTP servers** — optional on paper. Correlating logs across systems falls apart without accurate time.

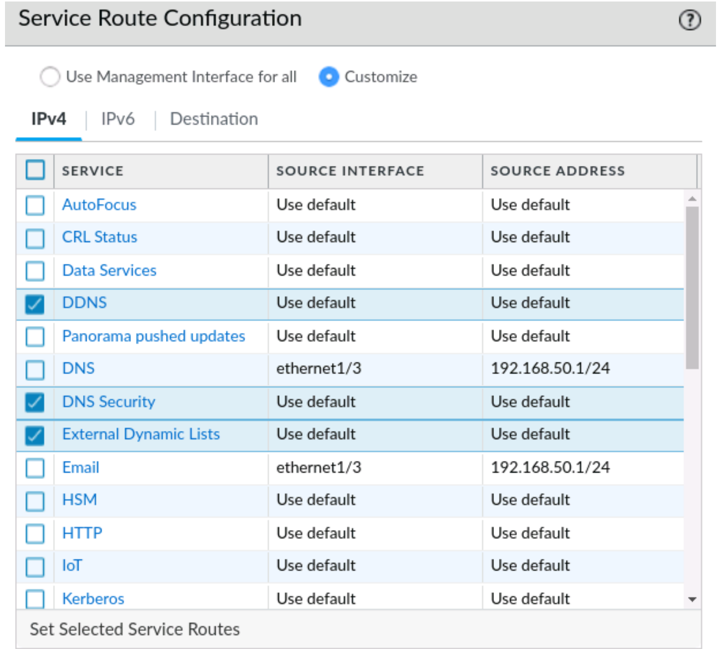




Service Routes

Device > Setup > Services > Service Route Configuration

- **Default behavior** — the MGT port reaches DNS, NTP, update, and authentication servers.
- **A service route** sends one named service out an in-band data plane port instead.
- **Why you would** — treat those flows like any other external traffic and inspect them before they reach the management plane.
- **Why you might not** — a true out-of-band management network already keeps the planes separate.



LESSON 4

Licenses, Software, and Content

Activate a Firewall

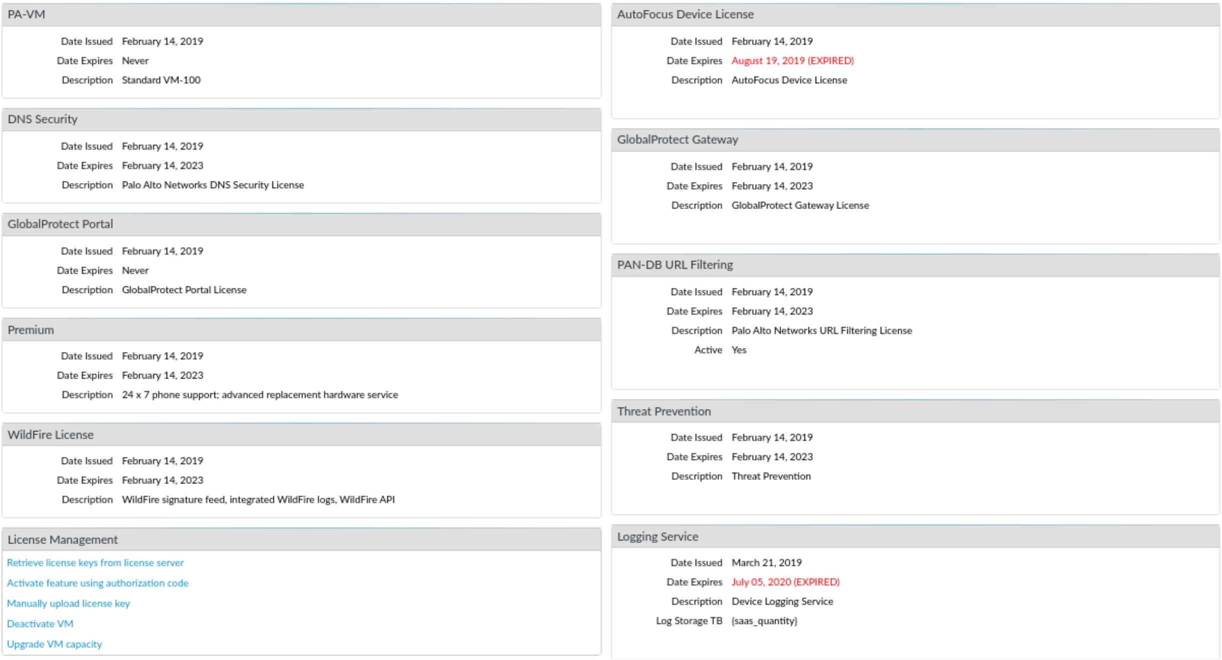
Registration and activation differ between hardware and virtual firewalls.

Step	Hardware firewall	VM-based firewall
Register with support	Serial number from the Dashboard	Emailed auth codes and order number
Activate licenses	Retrieve keys from the license server	Activate each feature by auth code
Verify servers	Update and DNS servers under Device > Setup > Services	Same
Content and software	Latest app and threat signatures, then the recommended PAN-OS release	Same

Licensing fails quietly and confusingly when DNS is wrong. Check Services first, every time.

Manage Licenses

- Every subscription appears at **Device > Licenses** with its issue and expiry dates.
- License Management links let you retrieve keys, activate by auth code, or upload a key manually.
- **Expired subscriptions** show in red — traffic keeps flowing, but that protection stops updating.
- Threat Prevention, WildFire, URL Filtering, DNS Security, and GlobalProtect are all licensed separately.



PAN-OS Software Updates

Device > Software — Check Now lists releases; download from the update server or upload an image, then install.

248 items

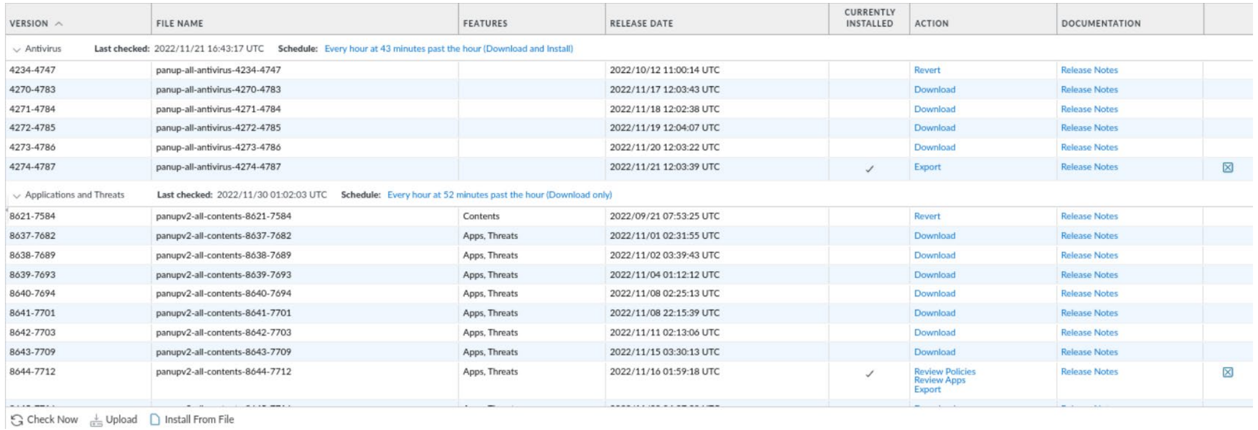
VERSION	SIZE	RELEASE DATE	AVAILABLE	CURRENTLY INSTALLED	ACTION		
11.0.0-c1361.main_rel	1025 MB	2022/07/13 18:09:48	Uploaded		Validate Export Install		
11.0.0-b4.dev_e_rel	1031 MB	2022/09/08 11:50:50	Uploaded		Validate Export Install		
11.0.0-b3.dev_e_rel	1006 MB	2022/08/17 16:54:18	Uploaded		Validate Export Install		
11.0.0	1037 MB	2022/11/17 08:45:28	Downloaded	✓	Validate Export Install Reinstall	Release Notes	
10.2.3	545 MB	2022/09/29 11:26:49			Validate Download	Release Notes	
10.2.2-h2	527 MB	2022/08/18 09:06:37			Validate Download	Release Notes	
10.2.2-h1	1010 MB	2022/06/28 10:12:10			Validate Download	Release Notes	
10.2.2	528 MB	2022/06/07 10:21:50			Validate Download	Release Notes	

Check Now Upload

Installing PAN-OS reboots the firewall. Plan the window; never do it because the list looked out of date.

Dynamic Updates

- **Device > Dynamic Updates** — antivirus, applications and threats, URL and DNS data.
- Schedule the **recurrence**, then choose download-only or download-and-install.
- Updates can be installed manually or **reverted** to a previous release.
- A content release can change how App-ID classifies traffic — which changes what your policy does.



VERSION	FILE NAME	FEATURES	RELEASE DATE	CURRENTLY INSTALLED	ACTION	DOCUMENTATION
Antivirus	Last checked: 2022/11/21 16:43:17 UTC	Schedule: Every hour at 43 minutes past the hour (Download and Install)				
4234-4747	panup-all-antivirus-4234-4747		2022/10/12 11:00:14 UTC		Revert	Release Notes
4270-4783	panup-all-antivirus-4270-4783		2022/11/17 12:03:43 UTC		Download	Release Notes
4271-4784	panup-all-antivirus-4271-4784		2022/11/18 12:02:38 UTC		Download	Release Notes
4272-4785	panup-all-antivirus-4272-4785		2022/11/19 12:04:07 UTC		Download	Release Notes
4273-4786	panup-all-antivirus-4273-4786		2022/11/20 12:03:22 UTC		Download	Release Notes
4274-4787	panup-all-antivirus-4274-4787		2022/11/21 12:03:39 UTC	✓	Export	Release Notes
Applications and Threats	Last checked: 2022/11/30 01:02:03 UTC	Schedule: Every hour at 52 minutes past the hour (Download only)				
8621-7584	panupv2-all-contents-8621-7584	Contents	2022/09/21 07:53:25 UTC		Revert	Release Notes
8637-7682	panupv2-all-contents-8637-7682	Apps, Threats	2022/11/01 02:31:55 UTC		Download	Release Notes
8638-7689	panupv2-all-contents-8638-7689	Apps, Threats	2022/11/02 03:39:43 UTC		Download	Release Notes
8639-7693	panupv2-all-contents-8639-7693	Apps, Threats	2022/11/04 01:12:12 UTC		Download	Release Notes
8640-7694	panupv2-all-contents-8640-7694	Apps, Threats	2022/11/08 02:25:13 UTC		Download	Release Notes
8641-7701	panupv2-all-contents-8641-7701	Apps, Threats	2022/11/08 22:15:39 UTC		Download	Release Notes
8642-7703	panupv2-all-contents-8642-7703	Apps, Threats	2022/11/11 02:13:06 UTC		Download	Release Notes
8643-7709	panupv2-all-contents-8643-7709	Apps, Threats	2022/11/15 03:30:13 UTC		Download	Release Notes
8644-7712	panupv2-all-contents-8644-7712	Apps, Threats	2022/11/16 01:59:18 UTC	✓	Review Policies Review Apps Export	Release Notes

Check Now   Upload   Install From File

Choosing an Update Posture

Palo Alto Networks frames content scheduling as a choice between two postures.

Security-first

- Protection outranks availability
- Check frequently, install with a short threshold
- Roughly six to twelve hours
- Accepts some risk of a policy surprise

Mission-critical

- Availability outranks immediate coverage
- Longer install threshold — commonly 24 to 48 hours
- Stagger rollout: low-risk sites first
- Accepts a longer window of exposure

Either way: Read the content release notes, review new and modified App-IDs for policy impact, and forward critical content alerts to whatever you monitor with.

The Security Thread in This Unit

One argument runs underneath every setting in this unit.

SECURITY FOCUS

An attacker who reaches the management interface does not have to evade your policy. They can rewrite it, disable your logging, and lock you out of your own firewall.

Never expose management to the internet or an untrusted zone. Restrict permitted IP addresses to bastion hosts. Allow only HTTPS and SSH. Enable only the services you actually use.

Why it matters: Every other control in this course assumes the firewall is still yours.

Data plane interfaces need the same treatment — that is what Interface Management Profiles are for.

Why This Is Not Theoretical

November 2024 — CVE-2024-0012, an authentication bypass in the PAN-OS management web interface.

- Unauthenticated attackers with network access to the **management web interface** gained administrator privileges.
- Chained with `CVE-2024-9474` for full takeover; added to CISA's Known Exploited Vulnerabilities catalog.
- Roughly **two thousand firewalls** were reported compromised.
- The vendor advisory: risk was **greatly reduced** for anyone who had restricted management access to trusted internal addresses.

The whole unit in one line

The Permitted IP Addresses entry you configure in today's lab is the control that made this a non-event for the organizations that had already set it.

What We Will Do Live

Firewall-A in NETLAB+ · web interface only · about fifteen minutes

- Log in and tour the tabs; find Commit and the Tasks queue.
- Load a named configuration snapshot and watch the job complete.
- Edit a field, show that traffic is unaffected, then commit it.
- Open Management Interface Settings and read the enabled services aloud.
- Add a permitted IP entry and discuss the mask we just wrote.
- Check for new PAN-OS software — and do not install it.

Take notes on the discussion, not the clicks. The clicks are in the lab guide.

Hands-On Lab: Configuring Initial Firewall Settings

NETLAB+ · Firewall-A at 192.168.1.254 from Client-A at 192.168.1.20 · credentials in the lab guide

- Connect to the web interface over HTTPS and load the baseline snapshot.
- Set primary and secondary DNS servers; verify the update server.
- Set primary and secondary NTP servers.
- Configure the domain, login banner, time zone, latitude, and longitude.
- Verify the MGT address and gateway, then add a permitted IP address.
- Commit, confirm the job in Task Manager, and check for new PAN-OS software.

The guide offers Detailed and High-Level steps. Use one section or the other — not both.

Check Your Understanding

Answer these before you start the lab.

- Which two paths allow initial configuration of a factory-default firewall?
- You changed a setting and closed the dialog. Is the firewall enforcing it? Why not?
- What does an empty Permitted IP Addresses list actually permit?
- Which is required for licensing to work — DNS or NTP? What breaks without the other?
- What is the difference between Device > Software and Device > Dynamic Updates?

If any of the five gave you trouble, review that lesson before starting the lab.

Key Takeaways

- Initial access is out of band — MGT or console, and the console is the last resort.
- Nothing you type is enforced until you commit. Candidate is not running.
- Restrict permitted IP addresses and allow only HTTPS and SSH on MGT.
- DNS is required; NTP is what makes your logs worth reading.
- PAN-OS updates replace the OS; dynamic updates refresh what it knows.

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Questions?

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